

UC San Diego

Rocket Propulsion Lab - Project Daedalus Mechanical Design, OpenRocket Simulation, Structural Analysis, and Subsystem Integration

Final Technical Report

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Design target: simulated apogee of approximately 4,086 ft using an AeroTech G80T motor

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1 Abstract

Project Daedalus was a UC San Diego Rocket Propulsion Laboratory student rocket built around a simulated 4,000-ft apogee objective. I served as Structures Lead, concentrating on nose-cone geometry, the shoulder/body interface, recovery attachment, structural CAD, manufacturability, and mechanical integration. The team manufactured and ground-tested the vehicle; it was never launched. The 4,086-ft apogee and 1.23-caliber stability margin are OpenRocket predictions, not measured flight results.

This report follows the complete vehicle from requirements through design and validation, then examines the recovery eyelet as a focused design iteration. The matched SolidWorks comparison uses an explicitly assumed 275 N bearing load, a 0.65 mm mesh target, and isotropic ABS to isolate the effect of eyelet geometry. It indicates 23.5% lower modeled peak stress and 20.3% lower modeled displacement for a wider rounded concept. Those results support a prototype decision; actual recovery dynamics, printed material behavior, and the connection to the nose cone remain unvalidated.

2 Introduction and Scope

A student rocket concentrates several coupled mechanical requirements into a small envelope. The nose establishes the aerodynamic profile, contributes to mass distribution, joins the body through a retention interface, and carries a recovery load path. Fins, motor hardware, parachute volume, wiring and electronics constrain the structural geometry around them. My responsibility was to make the structural interfaces fit, assemble and work with those neighboring systems.

The historical record establishes the design target, nominal dimensions, selected motor, design tools, manufacturing activity and ground testing. It does not supply as-built dimensional inspection, recovery force measurements, a launched flight, or a complete original eyelet drawing. Therefore the numerical load and comparative FEA in this report are engineering screens with labeled assumptions. The analysis is presented as a design evaluation, without implying that the wider concept was manufactured or flight qualified.

3 System Requirements and Vehicle-Level Targets

Quantity	Reference value	Engineering meaning
Vehicle envelope	25 in long; 1.5 in maximum diameter	Packaging, drag and printer constraints
Motor	AeroTech G80T; 133 N-s total impulse	Propulsion input to flight model
Mass with motor	1.038 lb / 0.471 kg	Modeled mass, not measured flight mass
Predicted apogee	4,086 ft / 1,245 m	Historical OpenRocket result; no flight
Predicted stability	1.23 calibers	Modeled CG/CP separation
Recovery	24 in parachute; nose separation	Drives fit and eyelet load path

The requirement was not altitude alone. A design that reached the simulated height but could not retain and separate the nose, support the recovery hardware, or package adjacent subsystems would not meet the system objective. For that reason, I treated the nose/body and recovery interfaces as vehicle-level decisions rather than isolated features.

4 Vehicle Architecture

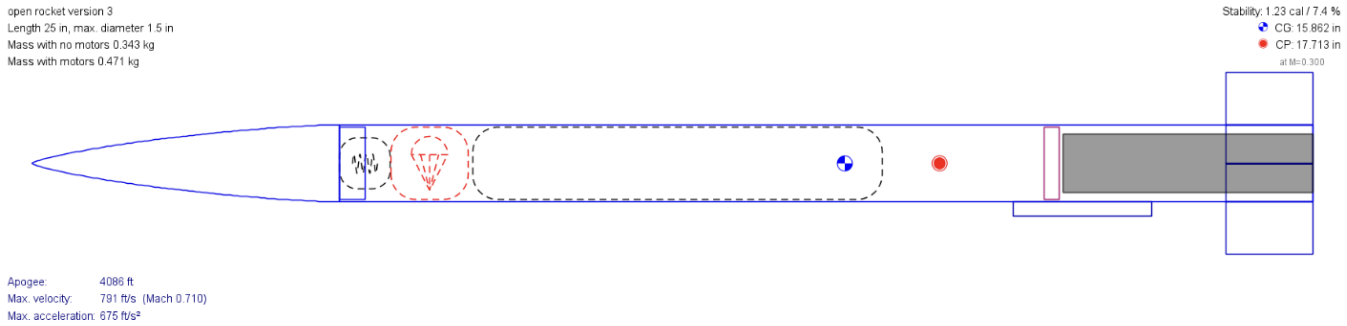


Figure 1. Historical OpenRocket vehicle model. The plot reports modeled CG and CP and a simulated 1.23-caliber stability margin.

Subsystem	Architecture and structural interface
Body	Approximately 19 in tube within the 25 in vehicle envelope; approximately 1.4 in bore.
Fin assembly	Four fins, about 1.7 in root chord and 1.02 in height, integrated with the lower body/fin can.
Recovery	24 in ripstop nylon parachute; nose separation; eyelet transmits line load into the nose.
Avionics	Arduino Pro Mini, BMP280, MPU6050, logger and LiPo constrained internal space; teammates owned electronics.
Motor	G80T motor and retention hardware determine lower-body packaging and axial load path.

My structural integration work addressed the physical consequences of these interfaces: sleeve and tube fit, internal access, mounting space, print envelope, recovery clearance and mass placement. The avionics package and propulsion design were team systems, not my sole design ownership.

5 Mechanical and Structural Design

The mechanical design cycle moved between the vehicle model and component CAD. A thicker wall or higher infill could increase local stiffness while changing forward mass and stability. A tighter shoulder could improve retention but make nose deployment uncertain. A larger eyelet ligament could lower stress but consume packaging volume and material. These are coupled trades, so I evaluated parts in the full-rocket context.

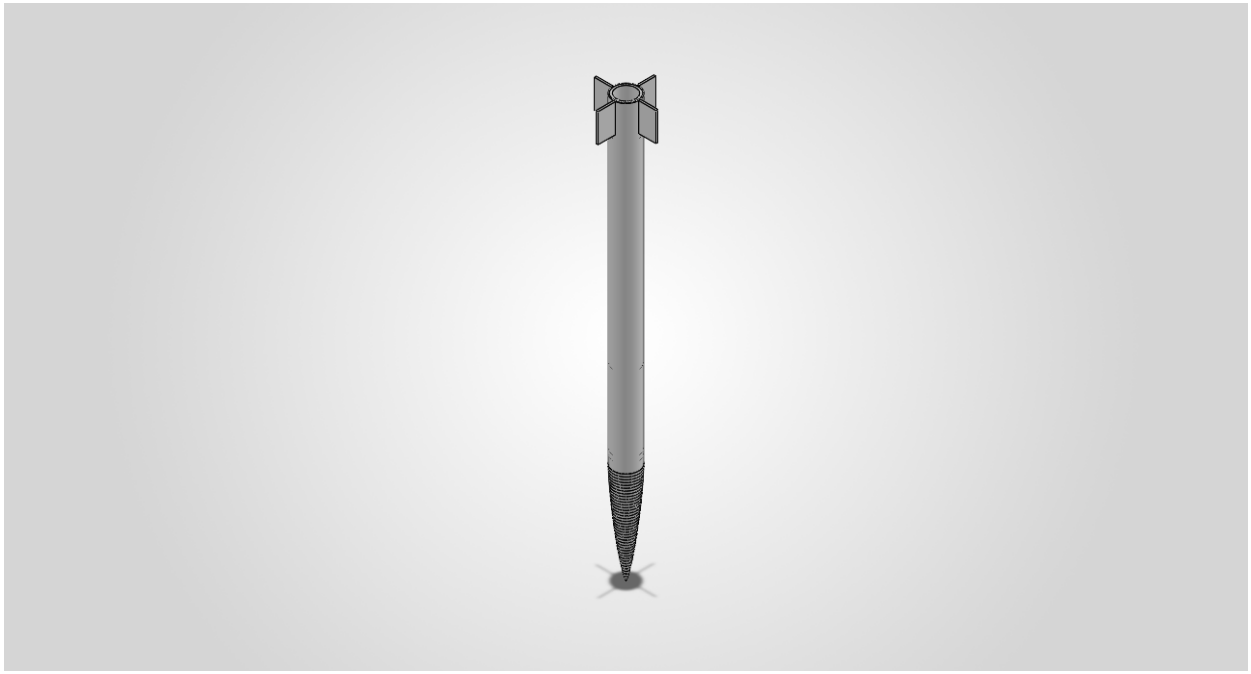


Figure 2. SolidWorks full-rocket envelope used to review relative geometry and packaging. It is a nominal CAD representation, not an as-built dimensional survey.

5.1 Structural load paths

The body and motor region carry thrust and inertial loads during ascent; the fin can carries aerodynamic fin loads into the tube. During recovery, parachute force travels through the shock cord and eyelet into the nose-cone structure, while the shoulder must release from the body. The comparative eyelet study isolates one link in that chain. It cannot establish the strength of the entire recovery system without the lug-to-nose connection and printed material being tested.

6 Nose-Cone Design

Parameter	Reference value	Constraint
Profile	Von Karman / Haack C = 0	Aerodynamic leading shape
Length and diameter	6.0 in; 1.5 in	Fineness ratio 4.0
Shoulder	1.400 in OD; 1.300 in ID; 0.5 in height	Retention, wall and print geometry
Manufacturing	In-house PLA printing; high infill	Stiffness, mass and print consistency

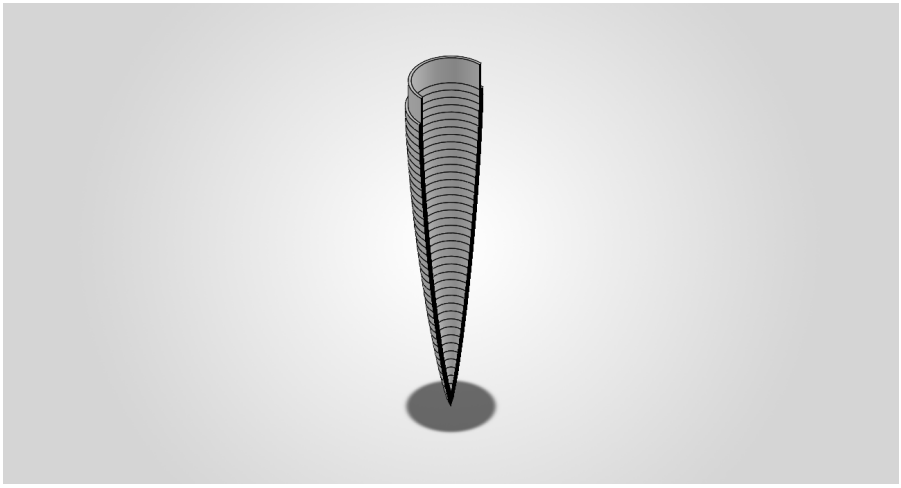


Figure 3. Native SolidWorks nose section; the displayed CAD should be read as nominal geometry. The exact manufactured bore and shoulder fit were not recorded.

$$r(x) = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \tfrac{1}{2}\sin(2\theta)}, \qquad \theta = \cos^{-1}\left(1 - \tfrac{2x}{L}\right)$$

Equation 1. Von Karman reference profile, with L = 6.0 in and R = 0.75 in.

The profile choice was bounded by printer volume, wall geometry, mass and the need to house the shoulder/recovery interface. A high infill setting increases local rigidity but does not by itself validate layer strength. Moving mass toward the nose also shifts the CG, so the OpenRocket model was needed to judge its vehicle-level effect.

7 Manufacturing and Tolerance Considerations

7.1 Shoulder fit

$$t_r = \frac{D_{OD} - D_{ID}}{2} = \frac{1.400 - 1.300}{2} = 0.050 \text{ in} = 1.27 \text{ mm}$$

Equation 2. Reference shoulder radial wall from the two listed diameters.

$$C_D = D_{body,\,ID} - D_{shoulder,\,OD} = 1.400 - 1.400 = 0.000 \text{ in}$$

Equation 3. The reference diameters give zero nominal diametral clearance; actual fit is unknown.

For a friction-fit starting point, I would finish the printed shoulder to approximately 0.001 in (0.0254 mm) larger than the measured tube bore, rather than assume the nominal CAD diameters control the fit. For a measured 1.400 in bore this gives an illustrative 1.401 in shoulder OD and 0.0505 in radial wall if the 1.300 in cavity is retained. This is an empirical finishing target, not an achieved hardware dimension.

$$F_{pull} \approx \mu\, p\, \pi D L$$

Equation 4. Pull force depends on friction, contact pressure, diameter and engagement length; nominal diameter alone cannot predict it.

7.2 Conditional retention and ejection checks

A screening calculation at the historical predicted peak speed of 241 m/s, with assumed sea-level density, drag coefficient 1.0, 0.300 kg coast mass and 0.050 kg nose mass, gives 35.6 kPa dynamic pressure and an approximate 6.76 N relative-separation load when local nose drag is neglected. I use this only to propose an acceptance test: 15-20 N measured axial extraction force, repeated across representative printed parts and conditions. The lower bound is 2.22 times the 6.76 N conditional screen.

$$R_{pull,\,min} \geq 15\ \text{N} > F_{sep,\,screen} = 6.76\ \text{N}$$

Equation 5. Proposed retention window, conditional on the screening assumptions.

A separate commanded-separation test should demonstrate at least 35 N of ejection force. Against the proposed 20 N upper pull limit plus 0.491 N for an assumed 50 g nose, the nominal remaining force is 14.51 N. No extraction force, ejection pressure, printed-part ovality, or thermal sensitivity has been measured in the supplied record. These numbers specify a test to perform; they are not historical test outcomes.

8 Avionics Packaging and Mechanical Interfaces

The avionics package set mechanical boundaries for internal diameter, access, mounting and wiring. I coordinated available structural space with teammates responsible for the Arduino Pro Mini, pressure and inertial sensors, SD logging and the LiPo battery. Their electrical design and package were not my individual deliverables. The mechanical requirement was to avoid a structural change that blocked wire routing, sensor access, removable hardware or deployment travel.

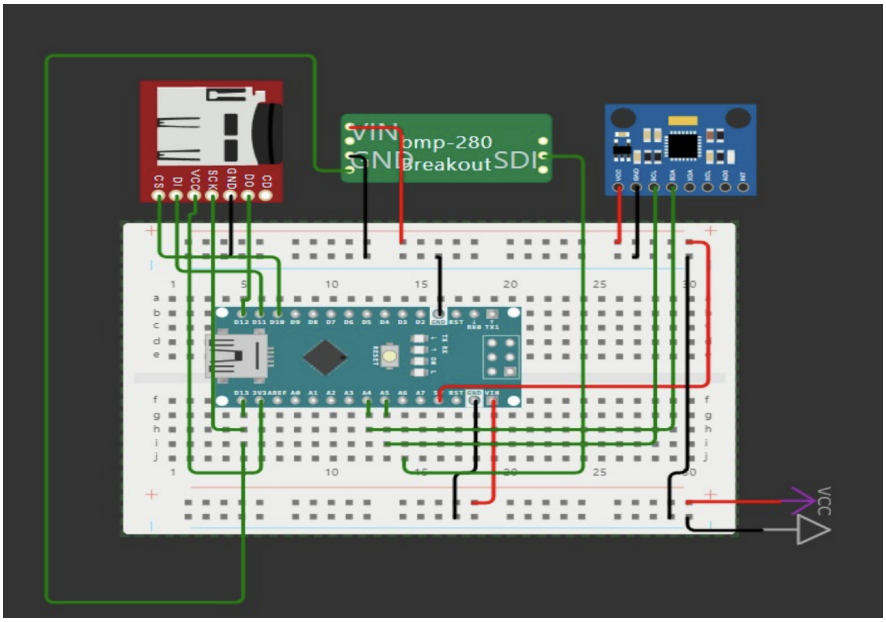


Figure 4. Early avionics architecture, included to explain mechanical packaging constraints rather than avionics design ownership.

9 OpenRocket Modeling and Predicted Flight Performance

OpenRocket supplied the historical performance estimate and a way to review how structural mass placement affects CG, CP, stability and drag. Its stored results are 4,086 ft apogee, 1.23 calibers stability, about 14.1 s to apogee, 791 ft/s peak velocity and 675 ft/s^2 peak acceleration. The original .ork file was unavailable for this report, so I do not claim an independently reproduced flight result.

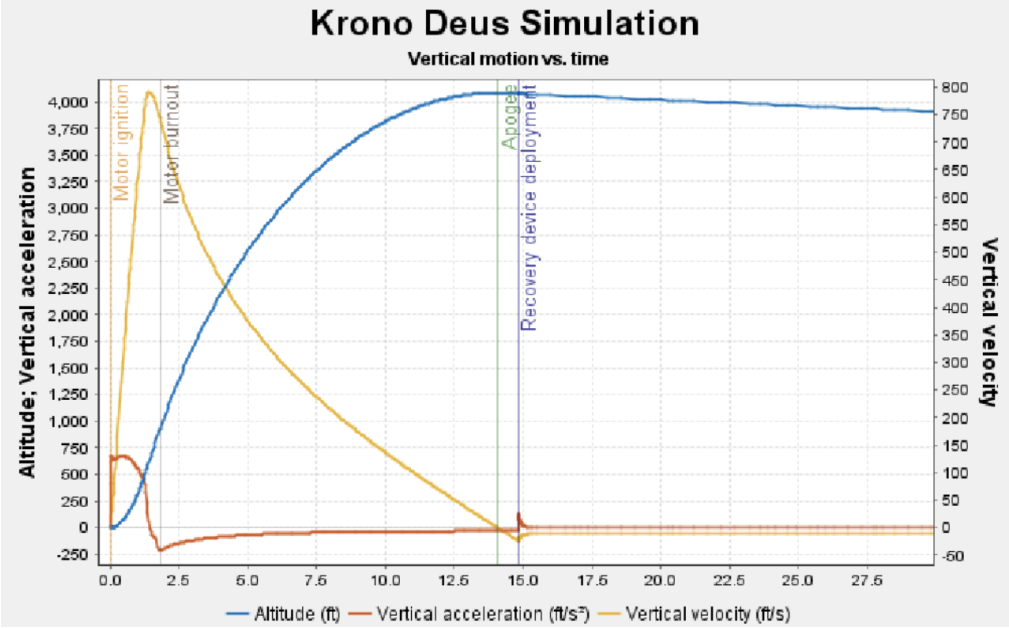


Figure 5. Historical simulated altitude, velocity and acceleration. The curves are predictions, not telemetry.

$$SM = \frac{x_{CP} - x_{CG}}{D} = 1.23 \text{ calibers} \quad \text{(historical model)}$$

Equation 6. Static margin definition; source positions were not independently remeasured.

A nominal thrust-to-weight check using the cited 78 N average thrust and 0.471 kg modeled mass gives 78/(0.471 x 9.81) = 16.9. This is a gross ratio from published model inputs, not a measured liftoff acceleration. Drag, motor thrust history and mass loss are needed for a flight acceleration profile.

10 Qualitative Flow and Pressure Review

An archived flow image offered qualitative feedback on external geometry. The available record does not identify the turbulence model, boundary conditions, mesh quality, convergence or experimental correlation, so the image cannot support a validated drag coefficient or pressure load. It remains useful as an early visualization of where the team examined flow behavior.

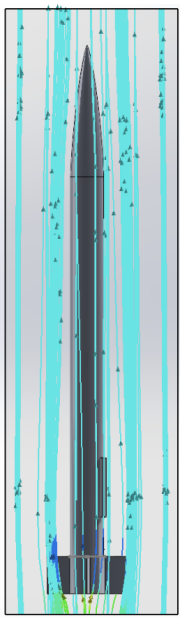


Figure 6. Archived flow/pressure visualization. Color trends are qualitative because the underlying simulation setup is incomplete.

11 Recovery Attachment: Load Path and Design Iteration

11.1 Free-body diagram and load selection

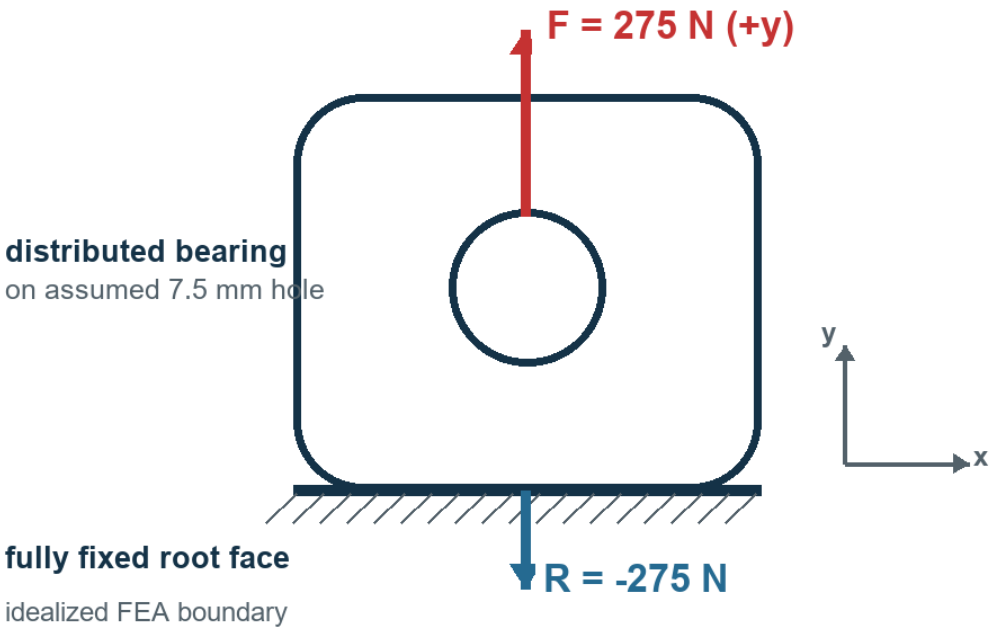


Figure 7. Eyelet free body: line load applied around the assumed hole, balanced by a root reaction. Real pin and nose contact are outside this model.

I selected a conditional 275 N static comparison case from an illustrative parachute-opening calculation. The assumed 24 in (0.6096 m) chute gives 0.29186 m^2 projected area. If deployment is delayed 1.5 s after apogee and the vehicle starts from rest, the simplified speed estimate is 14.715 m/s. At 1.225 kg/m^3 air density and assumed drag coefficient 1.75, steady drag is 67.74 N. An opening multiplier of 2 raises this to 135.48 N, and a separate structural screening multiplier of 2 gives 270.96 N, rounded to 275 N.

$$F_{screen} = K_s K_o \left(\frac{1}{2} \rho (gt)^2 C_D \frac{\pi D^2}{4} \right) = 270.96 \text{ N} \rightarrow 275 \text{ N}$$

Equation 7. Conditional design screen; neither multiplier is calibrated by a deployment measurement.

Assumed delay	Speed	Opening estimate	Structural screen
1.0 s	9.81 m/s	60.21 N	120.43 N
1.5 s	14.715 m/s	135.48 N	270.96 N
2.0 s	19.62 m/s	240.85 N	481.71 N

The 275 N selection covers the illustrative 1.5 s screen, not the 2.0 s case. It omits canopy inflation history, line stretch, body/nose mass split, deployment impulse, porosity, angle and air-density variation. Rocket mass does not enter this simplified drag expression. A flight load envelope requires measured or validated recovery dynamics.

11.2 Original-style versus wider rounded geometry



Figure 8. Native SolidWorks original-style rounded eyelet (left) and wider rounded concept (right). The original hole and root are assumed because a full manufacturing drawing was unavailable.

Feature	Original-style model	Wider concept
Envelope	17.78 x 17.78 x 12.7 mm	20 x 22 x 12.7 mm
Hole diameter	7.5 mm assumed	Same assumed 7.5 mm
Thickness	12.7 mm	12.7 mm
Reason for change	Reference curved profile	More net section and top ligament

The wider profile places material around the hole where load must flow toward the constrained root. Keeping hole, thickness, material and loading constant makes the contour comparison more interpretable. It also increases local material and envelope; the actual root/nose attachment may move the highest stress elsewhere.

$$p_{bearing,\,avg} = \frac{F}{td} = \frac{275}{12.7 \times 7.5} = 2.89 \text{ MPa}$$

Equation 8. Average projected bearing pressure, useful only as a first-order load check.

12 Structural Simulation and Mesh Verification

12.1 Matched SolidWorks setup

Model input	Both eyelet studies
Material	SolidWorks built-in isotropic ABS; no printed-part strength credit
Applied load	275 N distributed +Y bearing load on the assumed 7.5 mm hole
Fixture	Fully fixed root surface representing attachment restraint
Mesh	Matched 0.65 mm target; original 80,448 nodes; wider 110,803 nodes
Solver output	Von Mises stress, resultant displacement and reaction force

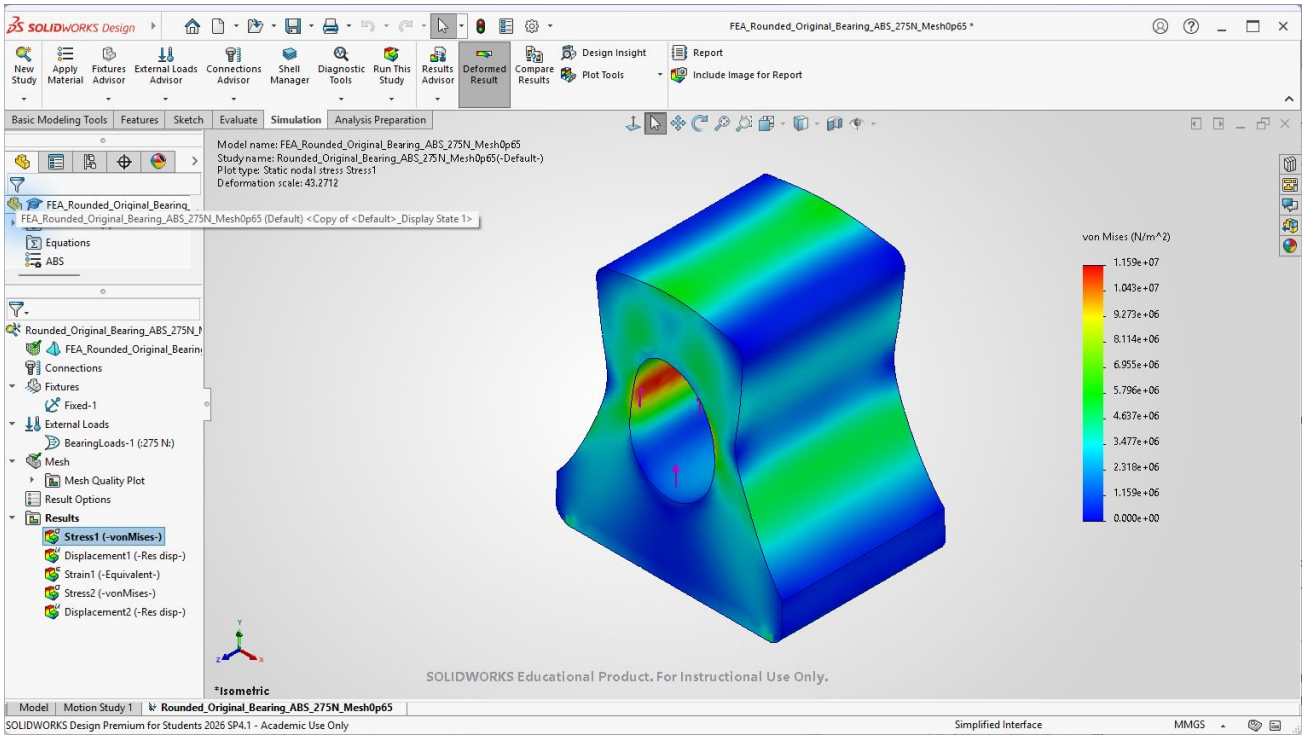


Figure 9a. Original-style eyelet, 275 N, 0.65 mm target: 11.591 MPa peak. Actual SolidWorks plot with load, fixture, legend and units.

The original-style model places the largest reported stress near the bearing hole and curved transition into the constrained root. The image retains the study tree, applied bearing arrows, root fixture and stress legend so the boundary conditions can be read with the result. A local contour maximum is sensitive to the idealized contact and root restraint; it is not a measured hardware stress.

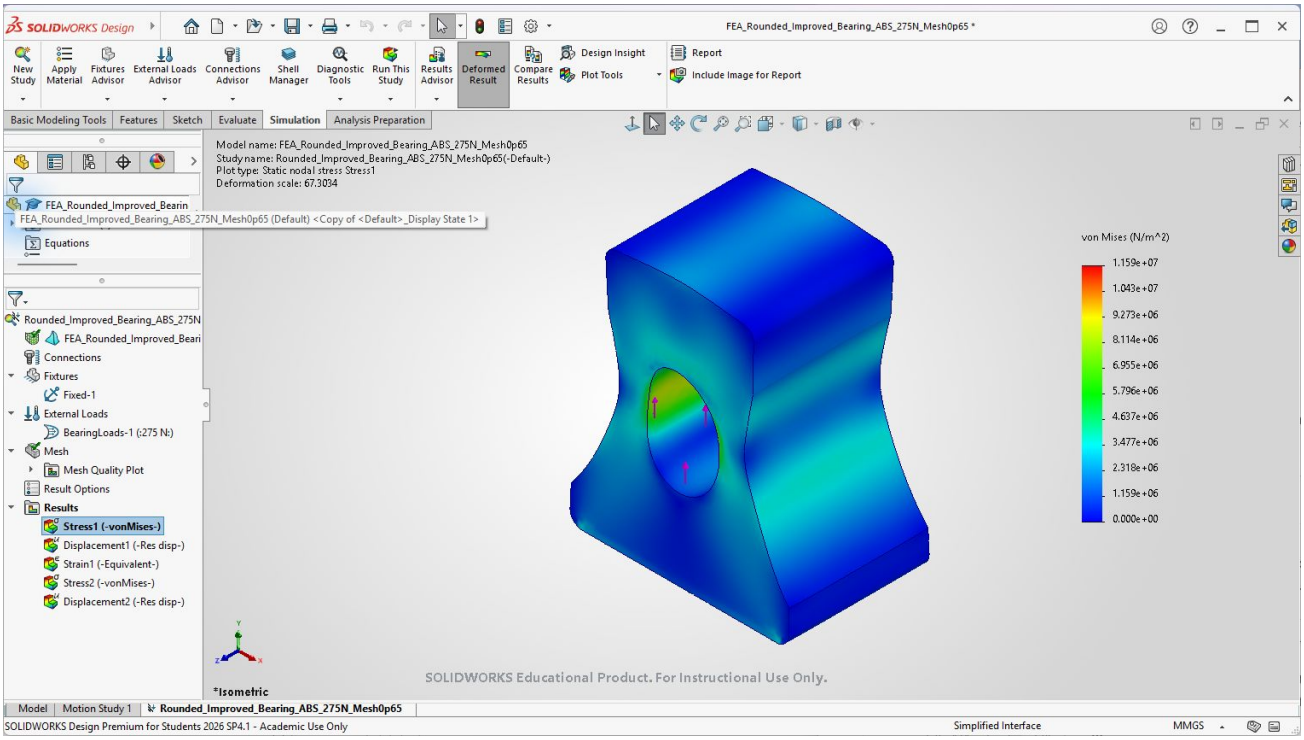


Figure 9b. Wider rounded concept, same orientation and common 0-11.591 MPa display scale: 8.869 MPa peak.

Output at 275 N	Original-style	Wider concept	Change
Peak von Mises	11.591 MPa	8.869 MPa	23.5% lower
Maximum displacement	0.04138 mm	0.03299 mm	20.3% lower
Root reaction Y	-275.0017 N	-274.9988 N	<0.002 N imbalance
Representative stress	2.846 MPa	1.335 MPa	Same extraction method

$$\Delta\sigma_{max} = \frac{11.591 - 8.869}{11.591} \times 100\% = 23.5\%$$

Equation 9. Modeled peak-stress reduction under matching conditional inputs.

$$\Delta u_{max} = \frac{0.04138 - 0.03299}{0.04138} \times 100\% = 20.3\%$$

Equation 10. Modeled maximum-displacement reduction under matching conditional inputs.

The common display scale makes the two stress fields visually comparable; the percentage changes come from saved solver values, not color matching. The lower peak and displacement are meaningful within the matched ABS model, but a material allowable and validated connection model are required before a safety factor or flight-ready claim can be made.

12.2 Mesh refinement and equilibrium

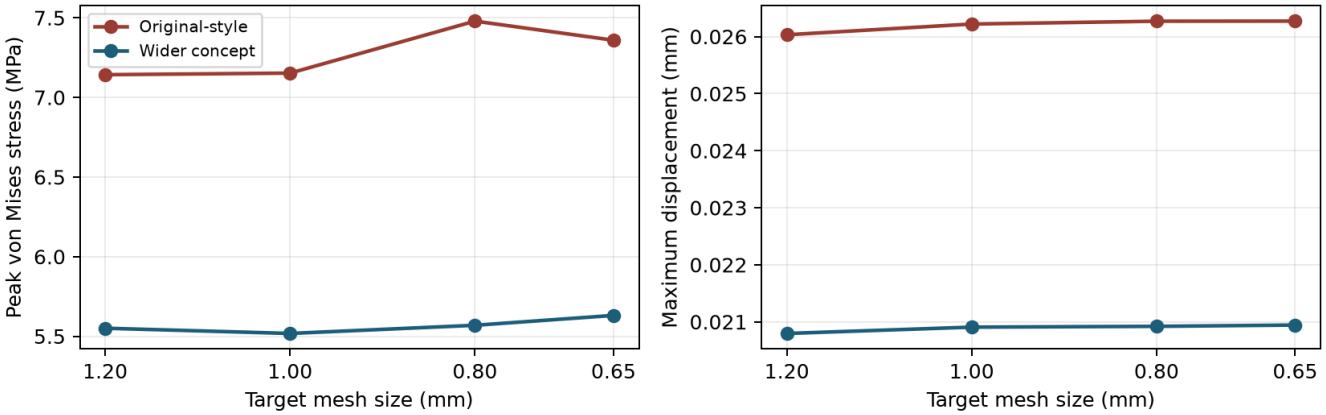


Figure 10. Existing four-level 174.59 N mesh series. Displacement approaches a stable value; peak stress fluctuates locally. The 275 N runs use the 0.65 mm target and are not a second four-level convergence study.

At the 0.65 mm target, the earlier 174.59 N series gave 7.359 MPa and 0.026271 mm for the original-style eyelet versus 5.631 MPa and 0.020944 mm for the wider concept. The later 275 N values scale by 275/174.5927 = 1.57509 as expected for this linear model. The normalized comparison is therefore internally consistent, but local peak stress did not vary monotonically with mesh. The fixed-root edge may introduce a stress concentration sensitive to idealization; a representative away-from-edge value should be reported alongside the peak.

The two 275 N reaction sums oppose the applied force within 0.002 N. This checks solver force equilibrium; it does not validate that the assumed load distribution, fixture, material or real load magnitude matches hardware. Native .SLDPRT, .CWR, .gsz, solver logs and full UI screenshots are retained with this report.

12.3 Material sensitivity

An assumed isotropic PLA run used E = 3.0 GPa, Poisson ratio 0.35, G = 1.111 GPa and density 1,240 kg/m^3 to test stiffness sensitivity. No tensile or interlayer allowable was entered. A print's raster, build orientation, porosity, temperature and layer bonding can dominate the actual failure path. Therefore this sensitivity cannot be translated into a printed-part factor of safety.

13 Design Process and Engineering Tradeoffs

13.1 From requirement to comparison

Step	Evidence and decision
Define	4,000-ft simulated target, vehicle envelope, stability, recovery and fabrication constraints.
Integrate	Nose, body, fin can, motor and avionics interfaces reviewed in vehicle context.
Check fit	Nominal zero-clearance shoulder stack led to a measured-bore finishing and force-test proposal.
Trace load	Recovery FBD made the hole, root, line direction and conditional 275 N screen explicit.
Compare	Matched rounded-eyelet ABS models show lower stress and displacement for the wider concept.
Test next	Measure printed fit, extraction/ejection, pin/nose connection and recovery dynamics.

13.2 What improved and what it costs

- Geometry: more top ligament and side width reduce modeled concentration but increase local mass and occupied space.

- Manufacturing: printed clearance and contact pressure determine the actual shoulder pull force; a nominal interference is only an initial finishing target.
- Model fidelity: a fully fixed root and distributed bearing force make a clean comparison, but a real pin, local contact and the nose attachment can change peak stress.
- System: added material shifts mass forward; retention and recovery loads must be evaluated without losing flight stability or deployment reliability.

14 Validation Status, Limitations, and Next Tests

Item	Status / limit
Manufacture and ground test	Team activity documented; no measured forces or inspection data supplied.
Launch and flight performance	No launch. OpenRocket predictions are not flight measurements.
Nose/body fit	No measured bore/shoulder dimensions or pull/ejection data; 0.001 in interference is proposed.
Eyelet CAD	Original-style hole and root dimensions assumed; wider version is a concept.
FEA	Saved solver results and UI screenshots verified for matched ABS comparison; root/real pin unvalidated.
Printed material	PLA stiffness sensitivity only; no build-direction or interlayer strength allowables.

Before flight use, I would inspect the actual body bore and printed shoulder; measure pull force over repeated assembly cycles and relevant temperatures; verify minimum ejection force; test the eyelet-to-nose joint and line/pin hardware; test printed coupons in the actual build orientation; and measure or bound parachute opening dynamics. A revised SolidWorks model should then include pin contact, the nose connection and off-axis loading. The current 275 N model is a design-screen comparison, not a released flight-hardware assessment.

15 Conclusion

Daedalus demonstrates the value of connecting CAD, flight modeling, manufacturing and recovery mechanics at the system level. My Structures Lead work centered on the nose-cone geometry and fit, recovery attachment, structural interfaces and making adjacent subsystems physically compatible. The team's manufactured and ground-tested rocket was never launched; the reported 4,086-ft altitude remains a historical prediction.

The focused eyelet iteration shows a technically useful trend: at the same assumed 275 N bearing load and 0.65 mm ABS mesh target, the wider rounded concept has 23.5% lower modeled peak stress and 20.3% lower displacement, with reaction forces in equilibrium. The defensible decision is to prototype and test it, while treating fit, printed material, pin contact, root attachment and recovery dynamics as open validation work.

Evidence and editable files

This report uses the archived project technical report, the final 275 N calculation record, native SolidWorks studies and screenshots, mesh-convergence CSV, and saved assembly/section images in FINAL_SUBMISSION. The editable builder for this PDF is build_daedalus_final_technical_report.py. Earlier report and case-study versions are preserved unchanged for traceability.